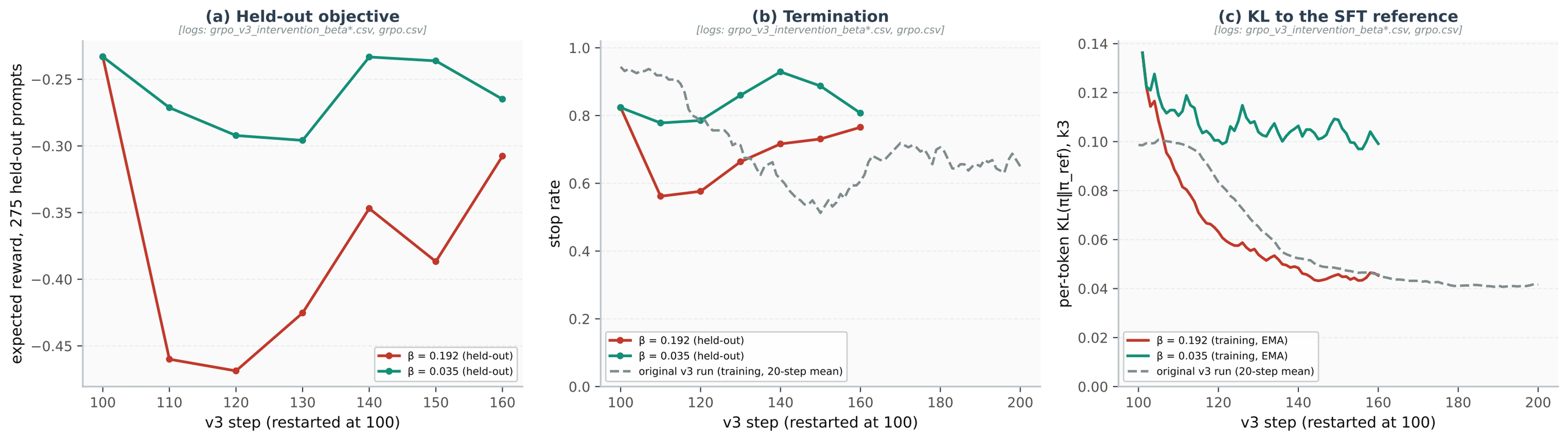
(a) Held-out objective
[logs: grpo_v3_intervention_beta*.csv, grpo.csv]
expected reward, 275 held-out prompts
−0.25
−0.30
−0.35
−0.40
−0.45
100
110
120
130
140
150
160
v3 step (restarted at 100)
β = 0.192 (held-out)
β = 0.035 (held-out)
(b) Termination
[logs: grpo_v3_intervention_beta*.csv, grpo.csv]
stop rate
1.0
0.8
0.6
0.4
0.2
0.0
100
120
140
160
180
200
v3 step (restarted at 100)
β = 0.192 (held-out)
β = 0.035 (held-out)
original v3 run (training, 20-step mean)
(c) KL to the SFT reference
[logs: grpo_v3_intervention_beta*.csv, grpo.csv]
per-token KL(π‖π_ref), k3
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
100
120
140
160
180
200
v3 step (restarted at 100)
β = 0.192 (training, EMA)
β = 0.035 (training, EMA)
original v3 run (20-step mean)